Before Convergence: A Formal Abstract-Art Interpretation of Deep Neural Network Loss Trajectories

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Abstract

The training process of deep neural networks is typically characterized by the Loss curve, whose dynamic changes reflect the optimization path and convergence state in the parameter space. Traditional studies mainly analyze the Loss curve from the perspectives of numerical optimization and generalization ability, while paying less attention to the visual features and structural tension presented by its formal structure. From a formalist perspective of abstract art, this paper analyzes the morphological characteristics of the Loss trajectory during the training process of deep neural networks. We point out that, prior to convergence, the Loss curve exhibits typical abstract formal features such as fragmentation, nonlinear oscillation, structural rupture, and staged reorganization. Through segmented analysis of trajectories at different training stages, this paper constructs a "formal mapping framework" that translates phenomena such as oscillations, plateaus, and abrupt changes in optimization dynamics into compositional elements in abstract art. The experimental analysis is based on typical training curve cases, verifying the explanatory power of this formal perspective for understanding the dynamic structure of Loss. This paper aims to provide an interdisciplinary formal analysis method, offering new visual and structural metaphors for understanding the training process of deep learning.

1 Introduction

The training process of deep neural networks is essentially a high-dimensional non-convex optimization problem, whose evolutionary trajectory is typically presented in the form of a curve showing the variation of Loss with respect to training steps. The Loss curve is widely used to evaluate model convergence speed, stability, and the risk of overfitting. However, this numerical tool also exhibits rich structural features at the visual level. Such features are particularly evident when network training fails to converge or collapses.

During the training process of deep neural networks, abnormal structures of the Loss curve can generally be summarized into three typical forms. The first type is violent oscillation. This phenomenon is manifested as large and frequent up-and-down fluctuations in the Loss value between adjacent iteration steps, with the overall trend lacking a stable downward direction. The second type is plateau stagnation. This phenomenon is manifested as the Loss value remaining relatively stable over a long time span, with the curve forming a horizontally extended structural segment along the time axis. The third type is abnormal increase. This phenomenon refers to sudden rebounds or sustained rising segments of the Loss value within an overall downward trend, breaking the continuity of the original convergence trajectory.

This structural process of the Loss curve, composed of oscillation, rupture, and reorganization, exhibits significant morphological similarity to the formal tension, fragmented composition, and nonlinear expression emphasized in abstract art. Specifically, the non-stationary fluctuations of the Loss curve along the temporal dimension form a highly deconstructed visual structure, in which the dense distribution of local extrema and frequent reversals of direction constitute formal characteristics similar to the "structural fragmentation" in abstract painting; staged plateaus and abrupt transition segments correspond to spatial stagnation and visual rupture in composition.

This paper does not discuss the historical context and symbolic meanings of abstract art, but focuses solely on its formal characteristics and compares them with the structural dynamics of the

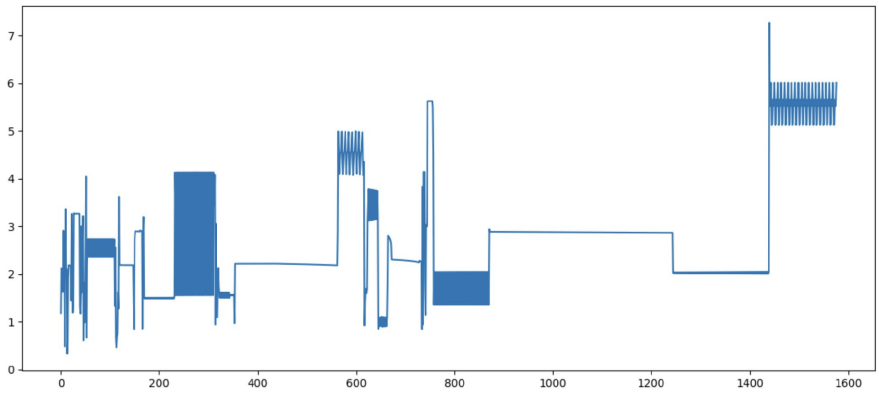

Figure 1: A Loss curve generated during an actual training process, with the horizontal axis representing training epochs and the vertical axis representing the Loss value.

Figure 2: The classic abstract painting Guernica.

Loss trajectory. Adopting a formalist analytical perspective, we use fragmented composition, nonlinear structure, and tension-generation mechanisms in abstract art as a reference framework to conduct structural decomposition and mapping analysis of the staged forms of the Loss curve during the training process of deep neural networks.

To enhance the concreteness and observability of the argument, this paper takes an actual training case as the research object and performs segmented extraction and morphological annotation of the Loss trajectory throughout its complete training cycle. In this case, we record the continuous dynamics of Loss values with respect to iteration steps, and structurally divide the oscillation intervals, plateau intervals, and direction-reversal intervals in the pre-convergence stage, thereby constructing correspondences at the formal level.

2 Case Study

Figure 1 presents the Loss curve generated during the training process of a real neural network, and Figure 2 presents the classic abstract painting Guernica. From the perspective of overall structure, the Loss curve does not exhibit a smooth monotonically decreasing form; rather, it is composed of multiple structurally distinct segments spliced together, forming a highly segmented dynamic composition, which creates a significant formal resonance with the cubist decomposition and spatial compression style commonly seen in abstract painting.

(1) Initial Stage: Fragmented High-Frequency Structure (approximately steps 0–150)In the early stage of training, the Loss curve exhibits intense high-frequency oscillations and frequent reversals of direction. The curve is dramatically stretched in the vertical direction and forms irregular polyline structures in the horizontal direction, with densely arranged local extrema that create a fragmented

visual rhythm. This form presents a pronounced sense of structural division and directional conflict, similar to abstract compositions in which continuous space is broken into multiple fragments coexisting in tension.

(2) Local Stacking and Block Compression (approximately steps 200–350)Within this interval, the Loss value oscillates densely around a certain range, forming a locally thick coverage band in two-dimensional space. Due to repeated numerical superposition, the line visually produces a "block-like" effect, reinforcing the volume and compression of the local structure. This trajectory organization, characterized by multiple overlaps and redrawings, bears strong similarity to the formal strategies in abstract painting that construct three-dimensional structures through line overlapping and spatial compression.

(3) Structural Rupture and Directional Reorganization (approximately steps 550–750)During this stage, multiple significant transitions and abrupt changes occur. The Loss value undergoes large jumps within short time spans, forming sharp inflection points and stretched segments. The overall direction is continuously reorganized across different intervals, disrupting the continuity of the original trend. Formally, this discontinuity and multi-directional juxtaposition create the effect of the time axis being cut and reassembled, enhancing the overall compositional tension and sense of instability.

(4) Extended Horizontal Composition (approximately steps 900–1200)The long plateau structure in this interval forms a horizontally extended stable segment. Compared with the previous high-frequency oscillations, this structure appears visually smoother; however, its extended duration creates a pronounced rhythmic contrast within the overall composition. This horizontally stretched compositional form often functions as a spatial stabilizing plane in abstract style, forming structural contrast with the surrounding ruptured structures.

(5) Periodic Oscillation and Terminal Intensification (after approximately step 1450)In the later stage of training, the Loss curve exhibits regularly repeated small-amplitude oscillations, combined with a noticeable vertical surge. This "repetition–intensification" rhythmic structure produces a final visual concentration before convergence. The periodic fluctuations introduce a sense of rhythm, while the abrupt change forms a structural emphasis point, resulting in a layered distribution of tension in the overall composition.

3 Conclusion

This paper analyzes the structural forms of the Loss trajectory during the training process of deep neural networks from a formalist perspective. Through a case study, we point out that, prior to convergence, the Loss curve exhibits characteristics such as fragmented segmentation, multidirectional tension, and structural reorganization, and that its mode of morphological organization shows significant similarity to the formal logic of abstract painting.

This paper does not attribute artistic meaning to the optimization process, but rather reveals the formal isomorphism between different systems at the structural level. This perspective provides a new descriptive language for understanding training dynamics and offers an interdisciplinary approach to the structural analysis of complex optimization processes.